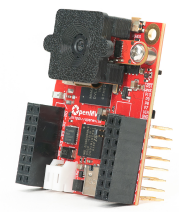
OpenMV® GENX320 Event Camera Module

Datasheet

SKU: OMV-MOD-GENX320-V2 / OMV-MOD-GENX320-V1

Description

The GENX320 Event Camera Module brings Prophesee's event-based vision sensing to the OpenMV Cam platform. Inspired by the human retina, each sensor pixel responds independently and asynchronously to relative changes in illumination, enabling efficient, high-speed, and low-power vision sensing. The sensor outputs 320x320 8-bit frames where pixel value 128 means no motion, values above 128 indicate increasing charge, and below 128 indicate decreasing charge. Also supports raw event mode for processing individual events with microsecond timestamps at 1\u20132 million events per second.



Target Areas

Event-based vision, high-speed motion detection, vibration analysis, gesture recognition, autonomous navigation, industrial inspection, low-power surveillance.

Key Features

- **375 Hz+ event histograms** — No exposure time, no motion blur. Configurable frame rate via sensor.set_framerate().
- **3 mW sensor power** — Ultra-low power — only scene changes are sensed.
- **140 dB dynamic range** — Sees changes in both dark and bright zones simultaneously.
- **<5 lux low-light** — Senses into twilight with 0.05 lux sensor cutoff.
- **320x320 output** — 8-bit grayscale event histogram frames or raw event stream at 1–2M events/sec.
- **Color palettes** — Software color palettes (EVT_DARK, EVT_LIGHT) for visualization.

Contents

1 Specifications

- 1.1 Sensor Specifications
- 1.2 Output Modes
- 1.3 Programming
- 1.4 Compatibility

2 General Info

3 Company Information

4 Reference Documentation

5 Revision History

1 Specifications

1.1 Sensor Specifications

Component	Details
Sensor	Prophesee GENX320 event vision sensor.
Resolution	320x320 pixels.
Event Rate	Event histograms at 375 Hz+. Configurable via sensor.set_framerate().
Power	Starting at 3 mW streaming data.
Dynamic Range	Up to 140 dB.
Low-Light	Senses below 5 lux; sensor cutoff at 0.05 lux.

1.2 Output Modes

Component	Details
Histogram Mode	320x320 8-bit frames. Pixel value 128 = no motion, >128 = charge increase, <128 = charge decrease. Events multiplied by 16 (default contrast). Adjustable via sensor.set_contrast() and sensor.set_brightness().
Event Mode	Raw event stream of 1024–65536 events per buffer. Each event contains type (PIX_ON, PIX_OFF, TRIGGER_HIGH, TRIGGER_LOW), microsecond timestamp, and x/y position. 1–2 million events per second throughput.
Color Palettes	image.PALETTE_EVT_DARK and image.PALETTE_EVT_LIGHT for visualization. Note: adds CPU overhead.

1.3 Programming

Component	Details
Pixel Format	sensor.set_pixformat(sensor.GRAYSCALE)
Frame Size	sensor.set_framesize(sensor.B320X320)
Examples	File > Examples > Cameras > Event Cameras > GenX320 in OpenMV IDE.

1.4 Compatibility

Component	Details
Supported Cameras	OpenMV Cam H7 Plus, OpenMV Cam RT1062, and OpenMV N6.

2 General Info

Dimension	Value
Weight	3 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/genx320-camera-module
Prophesee GENX320	https://www.prophesee.ai/event-based-sensor-genx320/
Prophesee Documentation	https://docs.prophesee.ai/stable/hw/partners/openmv.html
Focus Tool 3D Model	https://grabcad.com/library/genx320-module-cover-1
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.